

voi-rank v1: ranking scenarios by the decision value of a single measurement

voi-rank pipeline (protocol p001, run 7)

August 22, 2026

Abstract

We rank 68 scenarios by the economic value of one measurement sold to one decision-maker facing one binary decision. Each scenario is scored by the expected value of sample information (EVSI) of the measurement relative to its production cost, computed in closed form from seven parameters elicited by a small language model under a fixed, anchored protocol, with elicitation uncertainty propagated by Monte Carlo (100 000 draws). The primary metric is *efficiency* = EVSI/C : decision value per dollar of measurement. Because buyer counts and value capture are deliberately out of scope, the ranking measures per-measurement decision-value density, not business feasibility.

1 Introduction

A measurement is a product: someone with a decision to make pays for one observation because they expect to decide better with it than without it. This report treats that framing literally. The unit of analysis is a single *scenario*: one concrete decision-maker (an agent), one binary decision, one latent binary fact about the world, and one purchasable instrument that emits one binary signal about that fact. The canonical Bayesian decision problem then prices the information: how much better, in expected USD, does the agent do with the signal than without it?

That price is the expected value of sample information (EVSI). Dividing by the cost C of producing and delivering the measurement gives an efficiency: how many dollars of decision value one dollar of measurement buys. Ranking many scenarios by this efficiency is a way of asking: *where in the economy is a single observation worth the most, per dollar it costs to produce?*

Two design choices keep v1 honest and small. First, everything that would turn information value into business value — how many buyers exist, what fraction of the value a seller can capture, recurrence, rivalry — is deliberately out of scope; the interpretation caveat in Section 7 is load-bearing. Second, all seven parameters per scenario are elicited from a small language model under a fixed prompt protocol with worked anchor scenarios, repeated $k = 5$ times; the elicitation spread is propagated through every result. The pipeline is an experiment about LLM-based elicitation as much as a ranking.

2 Model

2.1 Structure

Latent state $\theta \in \{0, 1\}$, with the convention that $\theta = 1$ is the state in which responding is the correct action. The instrument produces a binary signal $x \in \{0, 1\}$. The agent then chooses an

action $a \in \{0, 1\}$ (respond / don't respond); the policy is derived by maximizing posterior expected utility, never elicited. Utilities are in the regret parameterization, $u(\theta, a=0) = 0$ for both states:

$$u(1, 1) = eB, \quad u(0, 1) = -K,$$

where $B > 0$ is the benefit of a correct response, $K > 0$ the net cost of a false-alarm response, and $e \in [0, 1]$ the action efficacy — the probability that the agent actually acts on the signal times the probability that the action delivers its benefit.

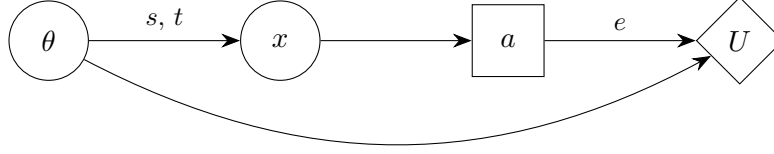


Figure 1: Influence diagram. The state θ (prior p) drives the signal x through sensitivity s and specificity t ; the agent observes x and chooses action a ; utility U depends on (θ, a) , with action efficacy e discounting the benefit edge.

2.2 Value of information in closed form

Write π for a belief $P(\theta=1)$. Acting optimally at belief π is worth

$$V(\pi) = \max(0, \pi eB - (1 - \pi)K),$$

the maximum over the two actions of posterior expected utility. With signal probabilities

$$P_1 = ps + (1 - p)(1 - t), \quad P_0 = 1 - P_1,$$

and posteriors $\pi_1 = ps/P_1$, $\pi_0 = p(1 - s)/P_0$ (a branch with zero probability contributes zero), the value of deciding *with* the signal is $P_1 V(\pi_1) + P_0 V(\pi_0)$, so

$$\text{EVSI} = P_1 V(\pi_1) + P_0 V(\pi_0) - V(p).$$

Perfect information replaces the signal with θ itself: the agent acts exactly when $\theta = 1$, which is worth $p eB + (1 - p) \cdot 0$. Hence

$$\text{EVPI} = p eB - V(p) = p eB - \max(0, p eB - (1 - p)K) = \min(p eB, (1 - p)K),$$

using $z - \max(0, z - w) = \min(z, w)$. The two arguments are the expected regrets of the two ignorant policies: never responding forgoes $p eB$; always responding costs $(1 - p)K$. Perfect information is worth exactly the smaller regret, and $0 \leq \text{EVSI} \leq \text{EVPI}$ always.

2.3 Per-scenario metrics

On every Monte Carlo draw we compute EVSI, EVPI, *efficiency* = EVSI/C (the primary ranking metric, ranked by median), *margin* = $\text{EVSI} - C$, and *headroom* = EVSI/EVPI where $\text{EVPI} > 0$ (how much of the perfect-information value this instrument delivers). Alongside the ranking we report $P_+ = P(\text{EVSI} > C)$ across draws: a high-efficiency scenario with low P_+ is fragile — its median hides a substantial probability that the measurement is not worth buying at all.

3 Method

3.1 Elicitation protocol

Each scenario’s seven parameters (p, s, t, e, B, K, C) are elicited from Claude Haiku via headless CLI calls (version 2.1.220 (Claude Code)), $k = 5$ independent repeats per scenario. Sampling temperature is not controllable from the CLI, so *repeats are the noise model*: cross-repeat spread is the empirical elicitation uncertainty. Calls run in an isolated context (no user configuration, no tools, a fixed system prompt), so the rendered prompt is the only input; its SHA-256 hash is stored with every response. The prompt (Appendix A) contains, in order: the model definition; two fully worked anchor scenarios with agreed numbers — an industrial vibration-survey decision and an everyday rapid strep test decision — to pin dollar and probability scales across scenarios; the scenario under elicitation; and per-parameter instructions requiring a short reasoning paragraph *before* the numbers. Each parameter is elicited as (p5, p50, p95) percentiles framed as “you would be surprised if the truth fell outside this interval.” Responses are validated (strict JSON, monotone percentiles, probabilities in $(0, 1)$, positive dollar amounts, an informativeness check $s_{50} > 1 - t_{50}$, non-degenerate prior); a failed attempt is retried once, then marked invalid. Validity across all attempts under the final protocol: 99.7% of 341 attempts. Total elicitation cost for the final protocol: \$17.35.

Scenarios themselves come from two sources: 8 hand-written seeds (doubling as smoke tests) and a proposer prompt that generates scenarios from ten seed domains under explicit constraints (one concrete decision-maker; θ a checkable binary fact; the instrument a purchasable measurement, not advice; stakes varied deliberately). No deduplication is applied: scenario text is its identity.

3.2 Percentiles to distributions

Dollar parameters (B, K, C) get lognormal fits; probability parameters (p, s, t, e) get Beta fits. Fits minimize squared quantile error against the three elicited percentiles (in log space for lognormals), initialized from moment matching. Every fit stores its parameters and residual in the database; badly asymmetric lognormal quantiles and Beta residuals above 0.02 raise a stored warning flag (89 warnings under the final protocol). Fits are never recomputed at analysis time.

3.3 Monte Carlo and sensitivity

Per scenario, 100 000 draws (seed 42) are taken with all seven parameters independent — a stated v1 assumption. Where a scenario has k valid repeats, each parameter draws from an equal-weight mixture over the k fitted distributions, so cross-repeat disagreement widens the intervals. Local sensitivity is the Spearman rank correlation between each parameter’s draws and the efficiency draws, per scenario. Global sensitivity aggregates $|\rho|$ over the top quartile of the ranking — the parameters that control the top of the list are where estimation effort should go next. Rank stability is computed from the aligned draws directly: the probability that a scenario ranks in the top 10 by efficiency across joint draws.

4 Scenario catalog

Table 1: Scenario catalog with pooled elicited medians (p50 across valid repeats). B, K, C in USD.

id	scenario	p	s	t	e	B	K	C
53	Online personal loan approval decision	0.65	0.85	0.95	0.78	1.8k	4k	15
60	Genomic risk stratification for acute leukemia transplant	0.32	0.94	0.95	0.55	400k	200k	2k
42	Industrial Gas Facility Safety Response	0.08	0.95	0.97	0.90	600k	150k	1k
41	Utility Transformer Replacement Decision	0.18	0.90	0.95	0.85	150k	90k	1.2k
5	Soil pathogen assay before planting a high-value crop	0.18	0.85	0.92	0.75	50k	10k	250
31	Grape maturity assessment for harvest timing	0.40	0.90	0.91	0.82	7k	3.5k	60
32	Seed germination quality control for cotton planting	0.12	0.90	0.94	0.80	10k	2.5k	60
46	Air Quality Health Alert	0.15	0.90	0.93	0.65	70k	12k	300
61	Serial troponin testing for acute coronary syndrome	0.15	0.95	0.92	0.90	100k	6k	400
52	Retail store credit card application approval	0.55	0.93	0.95	0.82	200	260	12
62	Pulmonary function testing for occupational lung disease	0.15	0.90	0.93	0.75	50k	20k	300
54	Residential mortgage approval for purchase	0.88	0.92	0.90	0.90	25k	60k	900
47	Brownfield Soil Remediation Decision	0.30	0.90	0.93	0.86	800k	300k	11k
48	Glacier Instability and Evacuation Risk	0.10	0.85	0.92	0.85	3M	400k	8k
13	Post-fire occupancy clearance decision	0.20	0.85	0.92	0.85	1M	100k	8k
40	Commercial Solar Panel Installation	0.45	0.90	0.92	0.80	100k	80k	5k
45	Water System Contamination Response	0.05	0.90	0.97	0.75	300k	150k	800
30	Bovine tuberculosis screening for dairy herd	0.06	0.88	0.95	0.75	80k	15k	800
22	Collaborative Robot Safety Deployment	0.68	0.88	0.90	0.80	200k	250k	10k
1	Vibration survey before pulling a critical gearbox	0.10	0.82	0.90	0.82	150k	15k	1.2k
59	Fractional flow reserve assessment for coronary intervention	0.50	0.93	0.94	0.75	25k	9k	1.5k
8	Counterparty credit report before extending 90-day trade credit	0.04	0.80	0.90	0.87	50k	10k	100

id	scenario	p	s	t	e	B	K	C
50	Landfill Methane Combustion Risk	0.12	0.84	0.90	0.80	650k	350k	10k
11	Soil bearing capacity design decision	0.18	0.85	0.87	0.88	1.5M	150k	20k
18	Autonomous Vehicle LiDAR Sensor Validation	0.72	0.88	0.90	0.82	250k	500k	30k
12	Hillside slope stability emergency response	0.15	0.85	0.88	0.82	1.5M	200k	28k
49	Ornamental Pond Water Quality Response	0.08	0.80	0.92	0.85	50k	5k	400
64	Furniture supplier transition decision	0.45	0.88	0.92	0.78	35k	40k	2k
28	Codling moth population assessment for apple orchard	0.28	0.82	0.92	0.75	9k	1.2k	200
27	Nitrogen deficiency detection in vegetable farm	0.27	0.88	0.92	0.68	3k	200	35
39	Residential Attic Insulation Investment	0.35	0.85	0.90	0.76	5k	2.8k	450
6	Structural inspection before purchasing a used mobile crane	0.10	0.85	0.92	0.82	50k	15k	2.5k
2	Pre-employment ergonomic screening before hiring a warehouse worker	0.12	0.75	0.85	0.65	15k	4k	250
35	Assembly Line Repetitive Strain Detection	0.15	0.88	0.92	0.70	70k	12k	4k
33	Warehouse Worker Back Injury Severity	0.25	0.80	0.85	0.78	5k	1.2k	250
4	Standardized manipulation benchmark of a warehouse robot before a flee	0.15	0.85	0.92	0.80	250k	150k	15k
9	Foundation crack severity assessment	0.20	0.85	0.88	0.80	15k	2k	1k
17	Container Ship Seaworthiness Certification	0.85	0.87	0.91	0.85	80k	1M	25k
34	Chemical Exposure Level Detection	0.22	0.85	0.92	0.72	40k	12k	2.8k
21	AI Model Partnership Evaluation	0.32	0.85	0.88	0.70	85k	65k	10k
65	Novel brake material production rollout decision	0.45	0.88	0.91	0.80	350k	500k	60k
57	H. pylori testing for chronic indigestion	0.22	0.94	0.96	0.80	650	250	140
24	Computer Vision Quality Inspection System	0.42	0.85	0.88	0.78	75k	65k	13k
51	Small-stakes fraud detection in wire transfer	0.78	0.87	0.90	0.88	100	160	2

id	scenario	p	s	t	e	B	K	C
66	Consumer electronics retail pilot program decision	0.32	0.77	0.82	0.80	200k	75k	20k
10	Window supplier quality verification	0.10	0.85	0.92	0.80	40k	12k	1.8k
58	Carotid ultrasound screening for statin initiation	0.20	0.85	0.90	0.65	9k	1.2k	550
29	Aquifer sustainability assessment for rice farming	0.30	0.80	0.85	0.72	90k	35k	7k
55	Syndicated corporate loan participation decision	0.80	0.89	0.88	0.80	1M	5M	150k
44	Preventive HVAC Filter Replacement	0.18	0.90	0.95	0.90	1k	150	180
38	Noise-Induced Hearing Loss Trend Detection	0.25	0.90	0.91	0.72	40k	6k	4.5k
20	Bicycle Brake System Safety Check	0.68	0.85	0.91	0.83	50	600	75
16	Vehicle Reliability for Evening Taxi Shift	0.78	0.85	0.88	0.82	200	320	100
43	Wind Farm Power Purchase Commitment	0.40	0.87	0.88	0.75	2M	1.5M	100k
23	Autonomous Vehicle Fleet Investment	0.40	0.80	0.85	0.72	250k	100k	50k
7	Sepsis-risk panel for a newly admitted patient	0.06	0.80	0.85	0.75	35k	1k	200
37	Heat Stroke Emergency Response Decision	0.18	0.82	0.85	0.88	200k	5k	150
68	Smartphone solid-state battery launch decision	0.50	0.85	0.88	0.82	50M	300M	250k
56	Catastrophic compliance and sanctions violation decision	0.08	0.80	0.88	0.80	400k	50k	500
14	Seismic retrofit investment decision	0.18	0.85	0.82	0.82	5M	8M	150k
63	Washing machine batch recall decision	0.15	0.85	0.92	0.85	500k	220k	3.5k
19	Track Safety Clearance Before Train Departure	0.88	0.90	0.90	0.90	15k	500k	1.5k
36	Head Injury Return-to-Work Clearance	0.20	0.83	0.88	0.72	35k	1k	350
3	Dangerous-capability evaluation of a frontier AI model before deployment	0.18	0.75	0.80	0.75	5M	1M	150k
67	Appliance rental fleet modernization decision	0.40	0.82	0.87	0.75	90k	60k	8.5k
26	AI Safety Measures Sufficiency Assessment	0.40	0.78	0.70	0.70	5M	1M	150k

id	scenario	p	s	t	e	B	K	C
15	Bridge Weight Capacity for Heavy Truck Routes	0.30	0.85	0.92	0.83	100k	1.5M	25k
25	AI-Assisted Surgical Robot Approval	0.40	0.85	0.84	0.70	450k	1M	50k

5 Results

Table 2: Final ranking by median efficiency (EVSI/C). P_+ = $P(\text{EVSI} > C)$ across draws; last column = parameter with largest $|\rho|$ vs efficiency.

rank	id	scenario	eff_{q05}	eff_{q50}	eff_{q95}	P_+	top
1	53	Online personal loan approval decision	0	25.7	261	0.89	C
2	60	Genomic risk stratification for acute leukemia transplant	1.25	19.6	79	0.95	B
3	42	Industrial Gas Facility Safety Response	0	18.2	175	0.87	B
4	41	Utility Transformer Replacement Decision	0	9.24	76.7	0.84	C
5	5	Soil pathogen assay before planting a high-value crop	0	6.91	53	0.78	B
6	31	Grape maturity assessment for harvest timing	0	5.84	42.7	0.79	K
7	32	Seed germination quality control for cotton planting	0	5.21	41.3	0.77	C
8	46	Air Quality Health Alert	0	5.05	148	0.67	C
9	61	Serial troponin testing for acute coronary syndrome	0	4.77	28.5	0.82	K
10	52	Retail store credit card application approval	0	4.35	42.6	0.80	C
11	62	Pulmonary function testing for occupational lung disease	0	3.77	53	0.67	B
12	54	Residential mortgage approval for purchase	0	3.67	17.8	0.77	K
13	47	Brownfield Soil Remediation Decision	0	3.53	24.8	0.75	C
14	48	Glacier Instability and Evacuation Risk	0	3.39	94.3	0.64	C
15	13	Post-fire occupancy clearance decision	0	2.91	29.1	0.65	K
16	40	Commercial Solar Panel Installation	0	2.85	16.7	0.75	C
17	45	Water System Contamination Response	0	2.08	125	0.54	B

rank	id	scenario	eff _{q05}	eff _{q50}	eff _{q95}	P_+	top
18	30	Bovine tuberculosis screening for dairy herd	0	1.81	13.1	0.63	p
19	22	Collaborative Robot Safety Deployment	0	1.78	13.1	0.63	K
20	1	Vibration survey before pulling a critical gearbox	0	1.74	17.6	0.60	C
21	59	Fractional flow reserve assessment for coronary intervention	0	1.64	10.1	0.66	K
22	8	Counterparty credit report before extending 90-day trade credit	0	1.62	80.1	0.52	B
23	50	Landfill Methane Combustion Risk	0	1.49	24.4	0.55	B
24	11	Soil bearing capacity design decision	0	1.32	16.8	0.55	K
25	18	Autonomous Vehicle LiDAR Sensor Validation	0	1.3	8.87	0.57	C

5.1 Reading the top of the ranking

The top decile of Table 2 is a recognizable economic type: a routine, commercially available instrument costing tens to a few thousand dollars — a credit-bureau inquiry, an employment verification, a laboratory assay, a sensor reading — attached to a decision whose stakes are two to five orders of magnitude larger, with an elicited prior in the middle band where neither acting nor not acting dominates by default. Four clusters recur: consumer credit decisions (an employment-verification check before a personal loan; a soft credit inquiry before a store card), acute-care medicine (genomic risk stratification before a transplant decision; serial troponin testing in chest-pain triage), agricultural quality control (soil pathogen assay, seed germination testing, grape maturity sampling), and industrial safety or asset decisions (laboratory gas analysis before an emergency response, load-profile analysis before replacing a transformer). The stored reasoning strings show the mechanism directly: the elicitor prices the instrument at its market rate, prices the decision’s dominant loss at a claim, outage, or crop scale, and lands on an efficiency far above one.

The ranking’s head deserves suspicion in proportion to its interval widths. The top-ranked scenario has $P_+ = 0.89$ but a q05 of zero, and its stored reasoning also illustrates a failure mode the validators cannot catch: in one repeat the benefit B is priced as the lender’s interest income from an approved loan rather than as the regret difference the model defines — a quiet re-anchoring of the utility convention. By contrast, the genomic risk-stratification scenario is the sturdiest entry in the list: it is the only scenario whose efficiency stays above 1 at its 5th percentile (Table 2), because both the stakes (transplant-or-not with survival differences priced in the hundreds of thousands) and the assay cost (a few thousand dollars) are comparatively well-pinned quantities.

Two structural observations from the bottom of the ranking balance the picture. First, 12 of 68 scenarios have median EVSI of zero, and the ones that stay dead across protocol variants share a signature: the prior already dictates the action (a track-safety check before a train departure happens regardless of any purchasable signal; a frontier-AI capability evaluation whose deployment-gating action is already optimal at the prior). In the model’s terms, information only has value when the decision sits near the indifference boundary; several intuitively “important” measurements are worthless per-decision because nobody’s action hinges on them. Second, scenarios whose single decision touches many people at once (a city air-quality alert) earn large B legitimately — one decision, population-scale consequences — which the per-measurement framing rewards without

any market-size unit error.

6 Sensitivity and elicitation quality

Global sensitivity over the top-quartile scenarios puts $|\rho|$ per parameter at: p : 0.20, s : 0.14, t : 0.13, e : 0.05, B : 0.33, K : 0.23, C : 0.40 — the ranking’s top is controlled chiefly by C , B , K . Cross-repeat noise over the final run’s $k = 5$ repeats (median over scenarios of the relative p50 spread; ranges over five repeats run higher than the matched- k comparison table in Section 6.1) is: p : 0.57, s : 0.06, t : 0.06, e : 0.14, B : 1.87, K : 1.80, C : 1.17.

6.1 Protocol comparison

Spearman rank correlations of median efficiency between the latest runs of each protocol (rows/columns are protocols; the manual column is the hand-entered seed baseline over its 8 scenarios, and p004 covers only the 17 deepened top-quartile scenarios):

	p000_manual	p001	p002	p003	p004
p000_manual	–	0.37	0.32	0.88	n/a
p001	0.37	–	0.58	0.46	0.69
p002	0.32	0.58	–	0.66	0.49
p003	0.88	0.46	0.66	–	0.50
p004	n/a	0.69	0.49	0.50	–

Median cross-repeat spread of the elicited p50, per parameter and protocol, computed over the first three repeats of each protocol so the comparison is at matched k (range statistics grow with the repeat count):

parameter	p001	p002	p003	p004
p	0.41	0.33	0.40	0.37
s	0.06	0.04	0.05	0.04
t	0.03	0.04	0.04	0.04
e	0.10	0.09	0.10	0.11
B	1.33	1.60	1.48	0.88
K	1.17	1.34	1.15	1.25
C	0.77	0.79	0.76	0.83

Three full-sweep protocol iterations were run before the final report run, each changing exactly one thing (Section 8). The pairwise rank correlations between full-sweep variants span $\rho = 0.46$ – 0.66 : minor prompt edits reshuffle the ranking substantially. The noise table explains why. The probability parameters s , t , e are elicited consistently (cross-repeat spreads of a few percent), the prior p moderately so, while the dollar parameters B , K , C carry relative spreads near or above 1.0 under *every* prompt variant tried — anchor-ratio cross-checks (p002) and forced component arithmetic (p003) both failed to reduce them. Efficiency is a ratio of exactly these noisy quantities, so the ranking inherits the noise. The correlations against the hand-entered seed baseline ($\rho = 0.37$ against the final run) swing widely across protocol variants in the matrix above: with only 8 shared scenarios they mostly measure small-sample noise, and cannot separate a model-vs-human gap from the prompt-vs-prompt gap. The practical reading of this report is therefore tier-based: membership

in the top decile, the broad middle, or the dead bottom is meaningful; adjacent ranks within a tier are not. The final run mitigates the repeat noise the only way that demonstrably worked — averaging over $k = 5$ repeats per scenario — and the interval plots carry the residual uncertainty honestly.

7 Limitations

Per-measurement value, not business value. With value capture and buyer counts removed by design, the ranking measures how much one observation is worth to one decision-maker per dollar it costs to produce. A top-ranked scenario is not a business case: it says nothing about how many such decisions exist, whether the decision-maker can be reached, or what fraction of the information value a seller could charge.

Independence. All seven parameters are drawn independently. In reality, e.g., high-stakes scenarios (B large) plausibly come with more expensive instruments (C large) and more careful priors; ignoring the correlations mostly widens intervals but can also bias efficiency medians.

The efficacy asymmetry. e multiplies only the benefit branch: $V(\pi) = \max(0, \pi e B - (1 - \pi)K)$. Since e includes the probability that the agent acts at all, a fully symmetric treatment would discount the false-alarm cost by the acting probability too; v1 keeps the simpler, conservative form (it understates the value of acting for noncompliant agents).

A small model as elicitor. Haiku is cheap enough to elicit hundreds of scenario-repeats, but small-model calibration is a real risk. The mitigations are the two anchors (scale pinning) and k repeats (noise measurement); Section 6 shows the resulting noise levels per parameter, and the protocol-comparison matrix shows how much the ranking moves when the prompt changes. Both are evidence about — not proof of — elicitation quality. The hand-entered seed baseline correlates with the final protocol’s ranking at $\rho = 0.37$ over 8 shared scenarios.

Selection effects of validation. The informativeness check ($s_{50} > 1 - t_{50}$) rejects elicitations rather than repairing them; a scenario whose instrument haiku consistently sees as uninformative drops out of the ranking instead of appearing with $\text{EVSI} \approx 0$.

Deferred by design. Capture fraction, number of buyers, annualization and recurrence, rivalry, compliance or stamp demand, risk aversion, parameter correlations, retrodiction validation, richer state and signal spaces (POMDPs), and deduplication are all out of scope for v1.

8 Learnings

The iteration loop (run a batch, compute health checks, change one thing, re-run, compare) was executed three times on the full scenario set, plus a sensitivity-guided deepening pass. What it taught, in decreasing order of generality:

Format compliance is a solved problem; scale calibration is not. Every batch cleared the validity gates almost perfectly (JSON validity $\geq 99.5\%$, constraint pass $\geq 99.5\%$, and 100% of slots valid after the single-retry rule) — the final protocol’s attempt-level validity of 99.7% saturates the

$\geq 95\%$ target with no tuning ever needed. All the difficulty lives in the numbers themselves: the dollar parameters came back with cross-repeat spreads near or above 1.0 in every protocol, while probabilities were reproducible to a few percent.

Two reasonable prompt fixes failed, informatively. Iteration 2 (p002) required an explicit anchor-ratio statement (“roughly $3\times$ the gearbox B ”) before each dollar percentile: dollar spreads got *worse*, plausibly because choosing an anchor and a ratio adds free variables instead of removing one. Iteration 3 (p003) required explicit component arithmetic (a Fermi decomposition summed in the reasoning): the model did the arithmetic faithfully, and the spread moved into the component values. Both changes were reverted. The residual dollar noise appears to be genuine epistemic uncertainty of the elicitor, not a formatting artifact — which is exactly what the repeats-as-noise-model design is meant to expose.

The ranking is noise-fragile at the rank level, stable at the tier level. Full-sweep variants correlate at $\rho = 0.46\text{--}0.66$; the rank-stability analysis of the final run (Figure 6) shows even top-3 scenarios are in the top 10 in well under all draws. Claims should be made about tiers, not adjacent ranks.

Deepening helped only marginally — and taught a methods lesson. The sensitivity analysis pinned the top of the ranking on C and B (Section 6), and a focused prompt variant (p004: vendor-invoice pricing for C , single-dominant-component sizing for B) was re-run on the top-quartile scenarios. At matched k , the targeted spreads improved modestly (C : $0.96 \rightarrow 0.83$; B : $1.00 \rightarrow 0.88$) while an untargeted parameter degraded by more (K : $0.80 \rightarrow 1.25$) — with 17 scenarios, indistinguishable from noise. A first comparison had looked dramatic and was an artifact of comparing 5-repeat ranges against 3-repeat ranges; repeat-noise comparisons must be made at matched k . The one intervention that reliably helped was buying more repeats: the final run pools $k = 5$.

Harness learnings. The headless CLI’s `--bare` isolation mode is incompatible with OAuth credentials in this environment; equivalent context isolation was achieved (and verified empirically) with disabled tools, disabled setting sources, and a fixed system prompt. The `--max-turns` flag of the spec no longer exists in the installed CLI; with tools disabled, calls are single-turn by construction. The model wrapped JSON in markdown fences from the very first smoke test, so fence-stripping before parsing is load-bearing. Storing the CLI’s full JSON envelope as the raw response preserved cost and session metadata without schema changes. An adversarial multi-agent review of the pipeline found (and led to fixes for) two real defects: a replay guard that missed added repeats, and a non-atomic elicitation write that could silently unbalance mixture weights after a crash.

For v2. Elicit dollar parameters on an explicit log-decade scale or with retrieval grounding; model parameter correlations (the independence assumption most distorts scenarios where stakes and costs co-vary); compare elicitors (a larger model as a second rater) on the same protocol; and validate the informativeness constraint’s rejection behavior against a signal-flip repair.

A Elicitation prompt template

The exact template of the final protocol (placeholders \$title, \$agent, \$decision, \$theta_definition, \$instrument are substituted per scenario):

```
You are performing a structured quantitative elicitation of the seven parameters of a
canonical Bayesian value-of-information model, for one specific scenario.

## The model

A single agent faces a single binary decision: respond (a=1) or do not respond (a=0).
The world is in a hidden binary state theta. Convention: theta=1 is the state in which
responding is the correct action.
The agent can buy exactly one measurement from an instrument. It produces a binary signal x,
where x=1 suggests theta=1.
The seven parameters:
- p: prior probability P(theta=1), given everything the agent already knows WITHOUT buying
this measurement. Base rates, visible symptoms, free records, common knowledge: all of that
is already inside p.
- s: sensitivity P(x=1 | theta=1) - the probability the instrument fires when the condition
is truly present.
- t: specificity P(x=0 | theta=0) - the probability the instrument stays silent when the
condition is truly absent.
- e: action efficacy = P(the agent actually responds after a positive signal) x P(the
response achieves its benefit). A number in [0,1].
- B: benefit of a correct response when theta=1, in USD, for this one decision. B = u(1,1) -
u(1,0): what responding gains over not responding when the condition is present.
- K: net cost of a false-alarm response when theta=0, in USD, for this one decision. K = u
(0,0) - u(0,1): what responding loses over not responding when the condition is absent.
- C: cost to produce and deliver ONE measurement, in USD.

All USD amounts are per single decision. Never use market sizes, annual totals, portfolio-
wide or fleet-wide numbers.

## Anchor scenarios (calibrate your scales against these agreed numbers)

Anchor A1. A plant reliability engineer decides whether to pull a critical gearbox for
overhaul. theta=1 = an incipient bearing fault is present. Instrument = one third-party
vibration-analysis survey.
- p: 0.02 / 0.08 / 0.25
- s: 0.60 / 0.80 / 0.95
- t: 0.70 / 0.90 / 0.98
- e: 0.50 / 0.80 / 0.95
- B (avoided unplanned failure minus planned repair): 20,000 / 150,000 / 1,500,000 USD
- K (unnecessary teardown + downtime): 2,000 / 10,000 / 60,000 USD
- C: 300 / 1,000 / 5,000 USD

Anchor A2. A primary-care physician decides whether to start a patient with acute sore
throat on antibiotics. theta=1 = group A streptococcal infection is present. Instrument =
one rapid antigen strep test.
- p: 0.05 / 0.15 / 0.35
- s: 0.70 / 0.85 / 0.95
- t: 0.90 / 0.95 / 0.99
- e (patient adheres x antibiotics deliver their benefit): 0.60 / 0.85 / 0.98
- B (avoided complications, shortened illness, reduced transmission): 50 / 300 / 3,000 USD
- K (unnecessary antibiotics: side effects, resistance contribution per decision): 20 / 100
/ 1,000 USD
- C: 5 / 15 / 50 USD

Each triple is p5 / p50 / p95 of the elicitor's belief. Note the scales: A1 is an industrial
six-figure-stakes decision, A2 is an everyday hundreds-of-USD decision. Place the scenario
below on a consistent scale relative to both.

## Scenario to elicit

- Title: $title
- Agent (the decision-maker): $agent
```

```

- Decision (respond vs not): $decision
- State definition: $theta_definition
- Instrument (the purchasable measurement): $instrument

## Instructions

For each of the seven parameters, in the order p, s, t, e, B, K, C:
1. First write a short paragraph of reasoning (2-4 sentences): what drives this quantity for
   THIS scenario, referencing base rates or typical magnitudes where you can.
2. Then commit to three percentiles p5, p50, p95. Read them as: p50 is your median estimate;
   you would be genuinely surprised if the true value fell outside [p5, p95]. They must be
   strictly increasing: p5 < p50 < p95.
3. State the unit ("probability" or "USD").

Remember for p: it must already reflect all information the agent has without buying this
measurement. If a free substitute for the instrument exists, its effect lives inside p, not
inside s or t.
Remember for B, K, C: single-decision USD amounts, consistent in scale with the anchors.

## Output

Output strict JSON only. No markdown fences, no prose outside the JSON. Exactly this shape,
with all seven parameters filled in:

{
  "parameters": {
    "p": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "s": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "t": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "e": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "B": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"},
    "K": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"},
    "C": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"}
  }
}

```

B Reproduction

Everything downstream of the elicitation database is deterministic given the stored seed (run 7: seed 42, 100 000 draws, code revision c14941af969d). The database stores every prompt hash, raw model response, fitted distribution, and run configuration; figures and tables in this report are generated exclusively from it. See `README.md` in the repository root for the run order.

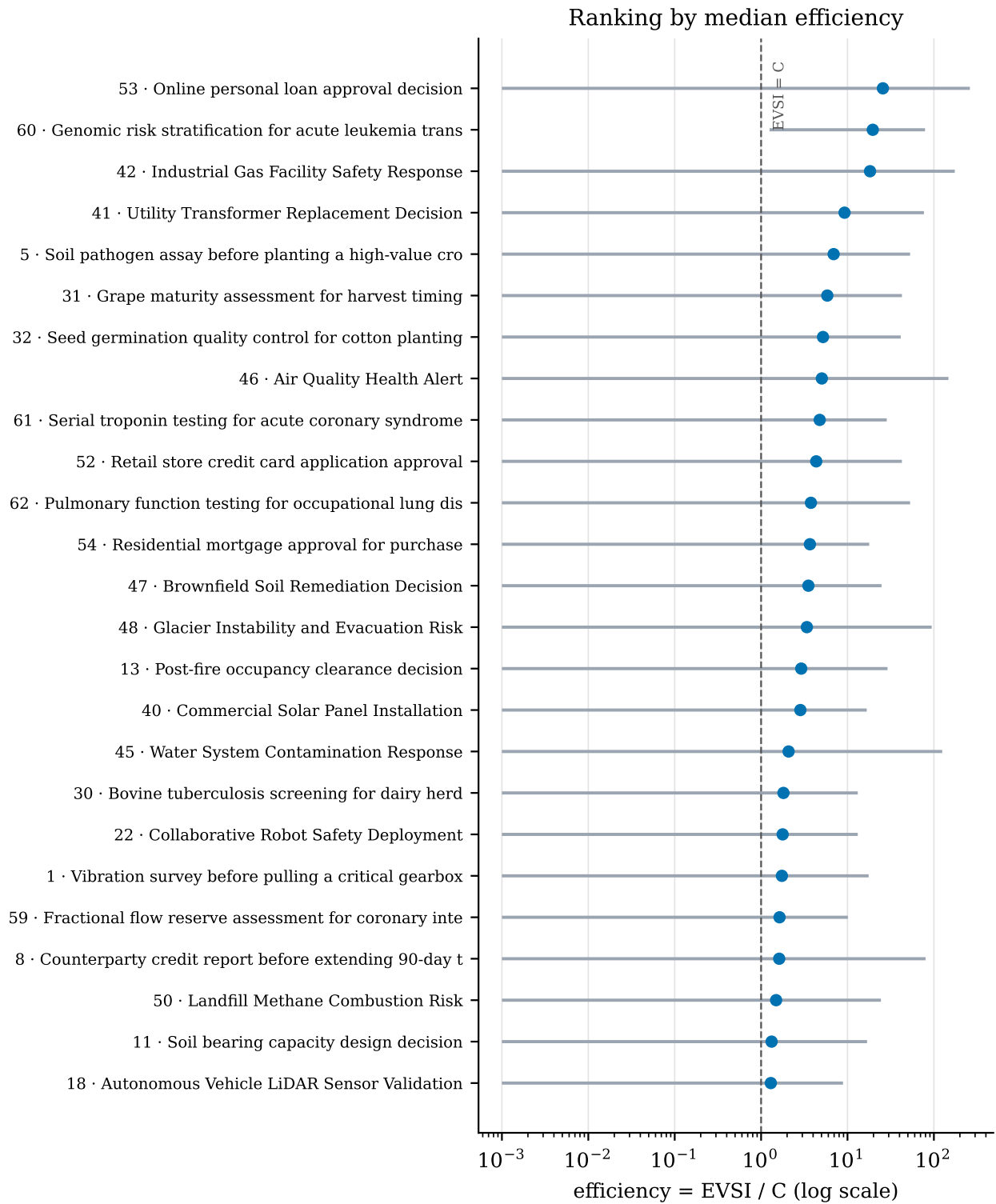


Figure 2: Top 25 scenarios by median efficiency $EVSI/C$. Intervals are q05–q95 across Monte Carlo draws; dots are medians (open dots: median 0, shown at the 10^{-3} axis floor). The dashed line marks $EVSI = C$: scenarios to its right buy more than a dollar of decision value per dollar of measurement.

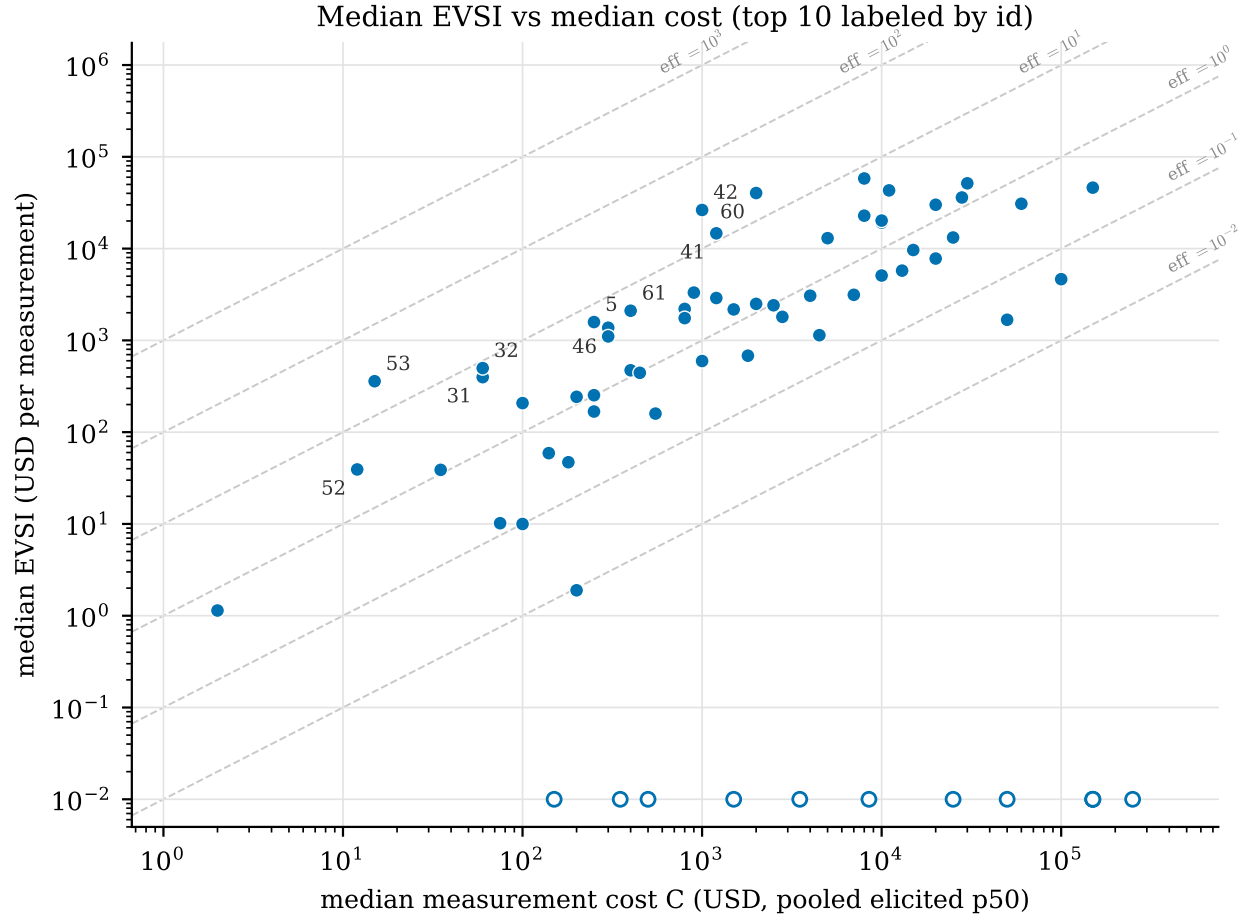


Figure 3: Median EVSI against median measurement cost, log-log, with iso-efficiency diagonals. The top 10 scenarios by median efficiency are labeled by catalog id. Open dots at the bottom edge are scenarios whose median EVSI is 0.

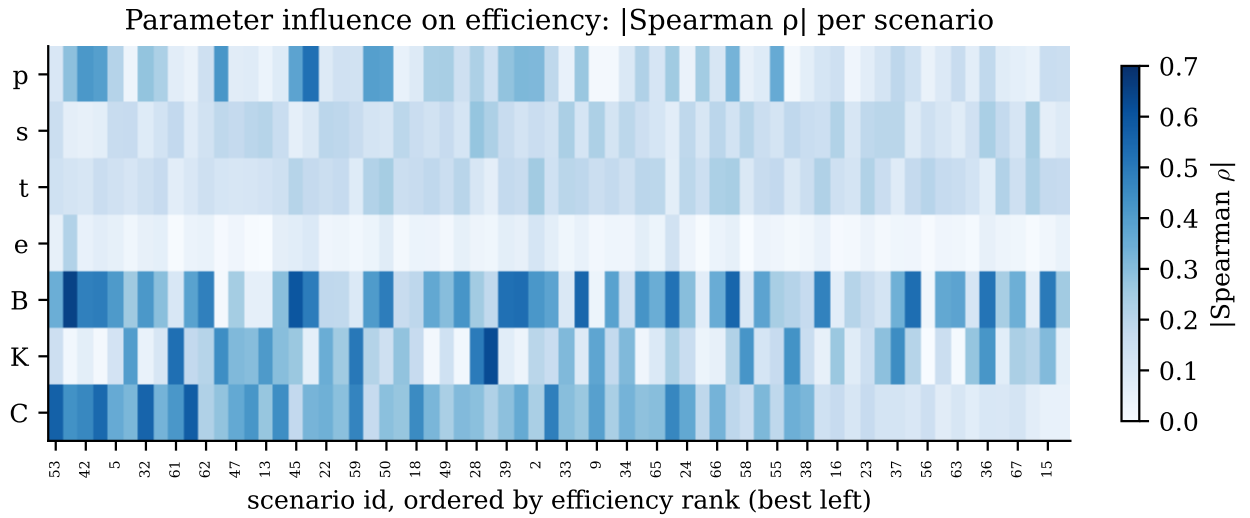


Figure 4: Absolute Spearman correlation between each parameter's draws and the efficiency draws, per scenario, scenarios ordered by rank.

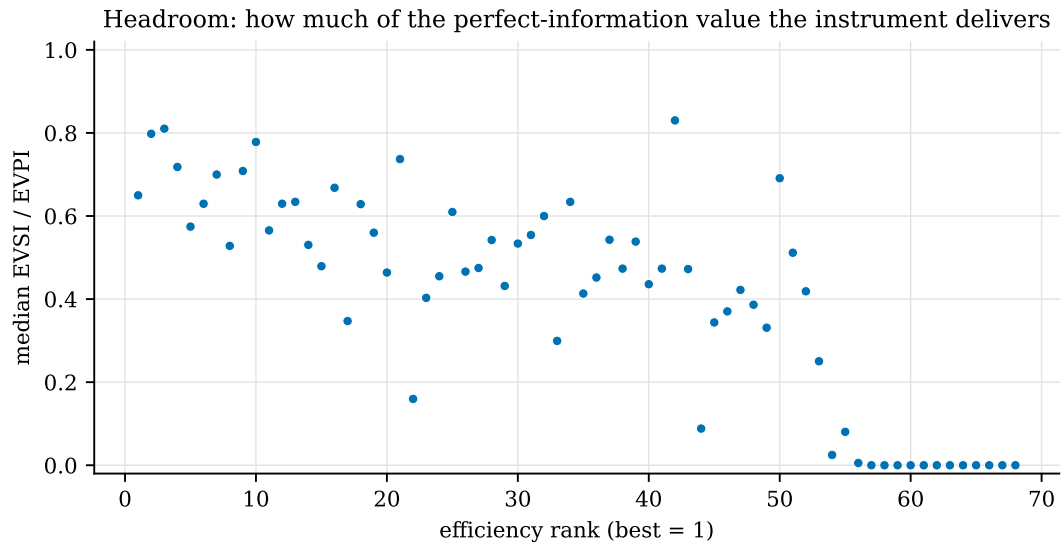


Figure 5: Median headroom EVSI/EVPI against rank. Low headroom at high rank marks decisions where a better instrument than the current one is worth building.

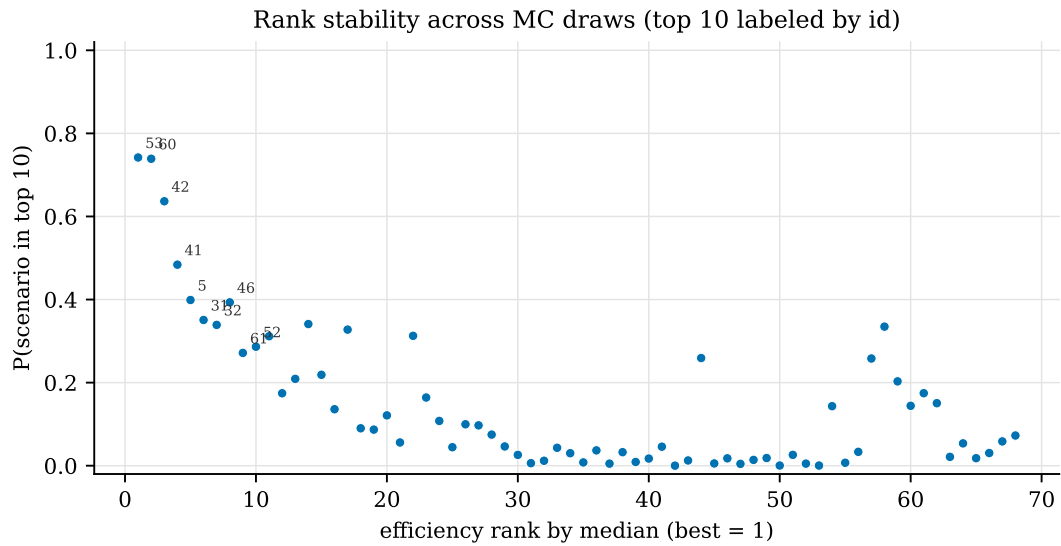


Figure 6: Rank stability: probability of membership in the top 10 by efficiency, across aligned Monte Carlo draws.

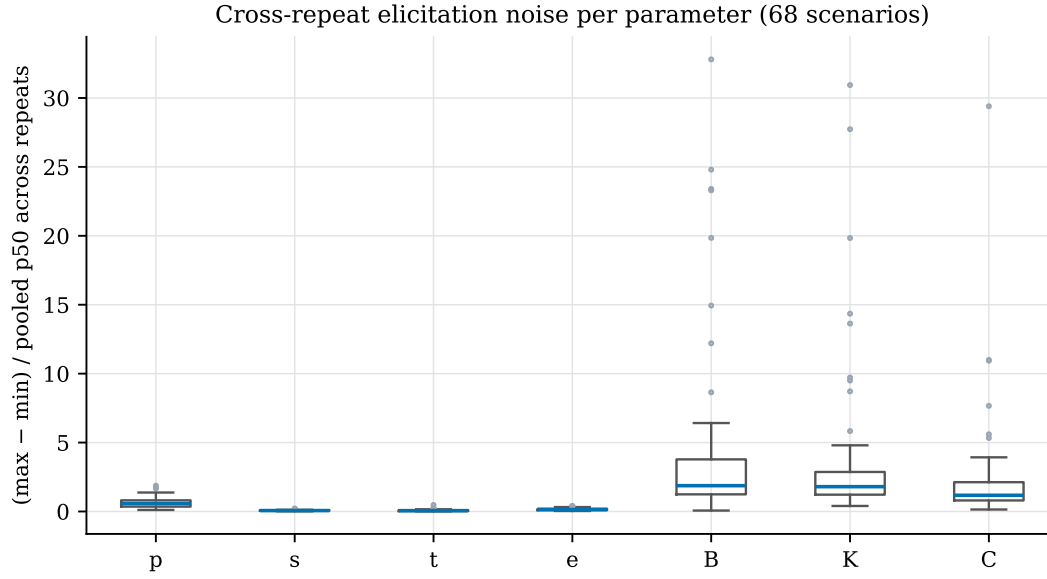


Figure 7: Cross-repeat elicitation noise: spread of the median estimate across $k = 5$ repeats, relative to the pooled median, per parameter.

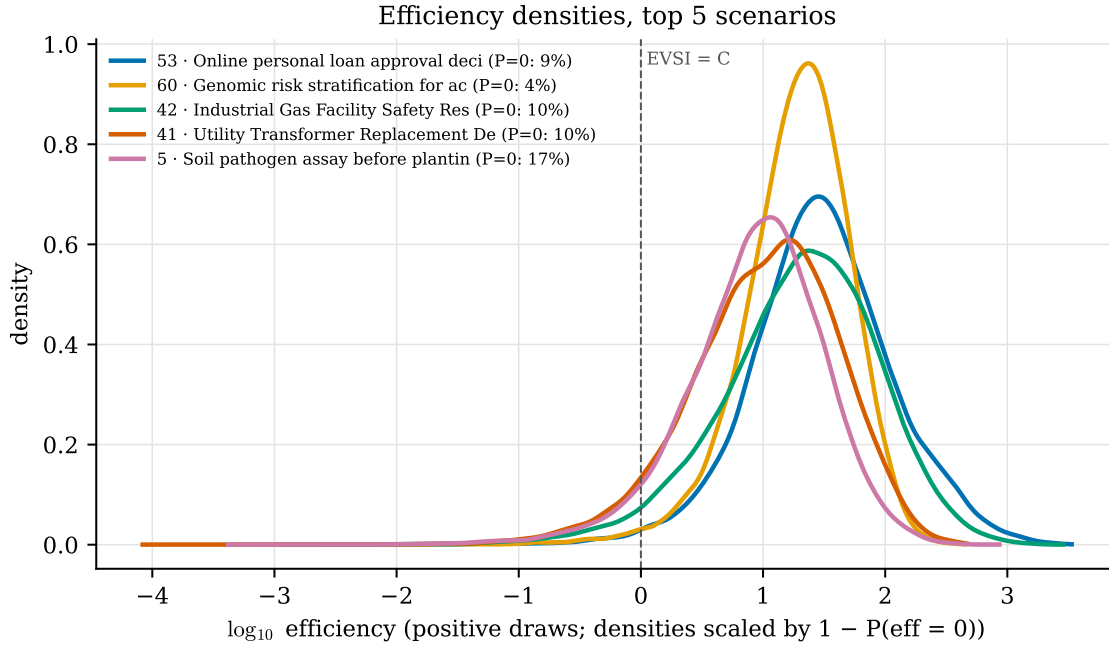


Figure 8: Efficiency densities of the top 5 scenarios over positive draws ($\log_{10}$ axis), scaled by $1 - P(\text{eff} = 0)$; the point mass at zero is the share of draws in which the signal never changes the decision.